



# Exercise 01: Simulate 2-to-1 mux

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- Set up your Ex01 folder
  - On a CAE machine, you should create an ece551 folder under your I:\ drive, and an Ex01 folder under that.
- Download provided files to your Ex01 folder
  - Canvas → In Class Exercises → Ex01 Files  
Download both **mux2to1.sv** and **mux2to1\_tb.sv**
- Look over both files to understand what's happening
  - mux2to1 implements a 2-to-1 mux using structural Verilog (see slide 43 in Lecture 1 for more info on structural Verilog)
  - mux2to1\_tb is a testbench for mux2to1. It instantiates the DUT and applies input stimuli (see slide 18 in Lecture 3 for more testbench info)

→ Once we know what to expect, let's simulate it in Modelsim/Questasim



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- Open Questasim
  - AppsAnywhere → Launch Quartus (we can close the window) → Windows search bar: type Questasim
- Create a new project
- Add the 2 files to the project
- Compile the code
- Start simulation on the testbench (mux2to1\_tb)
- Add waves of the DUT to the waveform viewer
- Run simulation
- Read the waveform & make sure it's what you expect
- Perform a screen capture of the waveforms (zoomed full)
- Submit the image to the dropbox for Exercise01